AAAI Press Formatting Instructions for Authors Using LaTeX — A Guide

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Abstract

Code — https://aaai.org/example/code
Datasets — https://aaai.org/example/datasets
Extended version —
https://aaai.org/example/extended-version

Introduction

In the course INFO-H410 Techniques of Artificial Intelligence at ULB, students are required to apply and compare different AI techniques to the same problem. The game chosen is 2048. 2048 is a single-player game on a 4×4 grid where the player has to combine tiles by shifting them in one of the four possible directions: up, down, left and right. Each turn, a new tile with a value of 2 or 4 appears in an empty spot on the board. The player wins by creating a tile with the value of 2048, but can continue playing to reach higher values. The game ends when there are no more valid moves available.
The randomness of the games makes it a good candidate for testing the performance of different AI techniques. Three agents will be compared; Deep qleaner, Expectimax and an n-tuple network TD learning agent. The goal will be to evaluate their performance (the final score reached) but also the computational cost to better understand the trade-offs between planning and leanred polcies.

Problem Formulation

The game 2048 is a sequential decision making problem with uncertainty. The state of the game is represented by the configuration of the board. The configuration of the board is a 4×4 grid where each cell can be empty or contain a tile with a value that is a power of 2. At each time step, the agent observes the current board configuration and selects in which direction to shift the tiles in one of the four possible directions: up, down, left and right as long as there is at least one tile moving. Two tiles with the same value that collide during a shift will merge into a tile with the value of their sum. A new tile with a value of 2 or 4 will then appear in a random empty spot on the board which makes future states only partially controllable by the agent. The objective of the agents is to achieve the highest score. The game ends when there are no more valid moves available. The fact that a new tile appears in a random empty spot on the board after each move makes the game stochastic and thus making it a good candidate for testing the performance of different AI techniques.

The action space is defined as $A = \{up, down, left, right\}$ where only moves that modify the board are considered valid. The state s corresponds to the values and positions of all tiles on the grid. Depending on the approach, the reward r can be constructed based on several features, such as the number of empty cell, monotonicity, placement of the largest tile, and score the sum of the values of the merged tiles. A terminal state is reached when no legal action is available.

As there is a large number of possible states and the game is stochastic, it makes exhaustive search or exact tabular (as in Q-learning) methods infeasible.

Methodologies

Three agents will be compared; Deep qleaner, Expectimax and an n-tuple network TD(0) learning agent. The three methods have each their own field of application and are based on different principles. Expectimax is a planning-based approach, deep Q-Learning is a deep reinforcement learning method that learns action values from experience meanwhile TD learning with function approximation is a lighter value-based alternative. Comparing them on the same problem will allow us to better understand the trade-offs between planning, learning, the computational cost as well as performance of each method. For each of the following approach, the states is encoded as a list of 16 integers, each representing the exponent k such that the tile value in that cell is 2^k. (For example, 16 is encoded as 4 since $2^4 = 16$

[*]With help from the AAAI Publications Committee.
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Algorithm 1: Training procedure of the TD agent

Require: Number of episodes N
Initialize TD agent with N-tuple network V_θ
for episode $= 1$ to N **do**
  Reset the environment
  Initialize empty trajectory path
  done $\leftarrow$ false
  while not done **do**
    Select an action using ϵ-greedy exploration
    Clone the environment and apply the action to obtain reward r
    Execute the action in the real environment
    Store the resulting after-state, reward, and terminal flag in the path
  end while
  for each transition $(s, r, s', done)$ in the trajectory path **do**
    Compute $v \leftarrow V_\theta(s)$
    if done **then**
      $\delta \leftarrow r - v$
    else
      $\delta \leftarrow r + V_\theta(s') - v$
    end if
    Update the weights of V_θ using δ
  end for
  Update $\epsilon \leftarrow \max(\epsilon_{\min}, \epsilon \cdot \lambda)$
end forreturn trained agent

with the particularity that 0 represents an empty cell). The action space is the same for all agents and consists of the 4 possible moves: up, down, left and right encoded as integer values from 0 to 3.

Expectimax

Deep qleaner

N-tuple network TD(0) learning agent

Temporal-Difference learning with value function approximation provides a lighter alternative to Deep Q-Learning. Instead of learning action values directly, this approach estimates the value of board after-states and updates this estimate by bootstrapping from successor after-states. In our implementation, the value function is approximated using an N-tuple network composed of lookup tables defined over several tile patterns and their symmetric variants. LaTeX-This representation is well suited to 2048 because it captures local board structures while remaining significantly cheaper than deep neural approaches. Compared with Deep Q-Learning, this method is simpler and more computationally efficient, although its performance depends strongly on the choice of patterns and representation.

Using LaTeX to Format Your Paper

The latest version of the AAAI style file is available on AAAI's website. Download this file and place it in the TeX search path. Placing it in the same directory as the paper should also work. You must download the latest version of the complete AAAI Author Kit so that you will have the latest instruction set and style file.

Document Preamble

In the LaTeX source for your paper, you **must** place the following lines as shown in the example in this subsection. This command set-up is for three authors. Add or subtract author and address lines as necessary, and uncomment the portions that apply to you. In most instances, this is all you need to do to format your paper in the Times font. The helvet package will cause Helvetica to be used for sans serif. These files are part of the PSNFSS2e package, which is freely available from many Internet sites (and is often part of a standard installation).

Leave the setcounter for section number depth commented out and set at 0 unless you want to add section numbers to your paper. If you do add section numbers, you must uncomment this line and change the number to 1 (for section numbers), or 2 (for section and subsection numbers). The style file will not work properly with numbering of subsubsections, so do not use a number higher than 2.

The Following Must Appear in Your Preamble

```
\documentclass[letterpaper]{article}
% DO NOT CHANGE THIS
\usepackage{aaai2026} % DO NOT CHANGE THIS
\usepackage{times} % DO NOT CHANGE THIS
\usepackage{helvet} % DO NOT CHANGE THIS
\usepackage{courier} % DO NOT CHANGE THIS
\usepackage[hyphens]{url} % DO NOT CHANGE THIS
\usepackage{graphicx} % DO NOT CHANGE THIS
\urlstyle{rm} % DO NOT CHANGE THIS
\def\UrlFont{\rm} % DO NOT CHANGE THIS
\usepackage{graphicx}  % DO NOT CHANGE THIS
\usepackage{natbib}  % DO NOT CHANGE THIS
\usepackage{caption}  % DO NOT CHANGE THIS
\frenchspacing % DO NOT CHANGE THIS
\setlength{\pdfpagewidth}{8.5in} % DO NOT CHANGE THIS
\setlength{\pdfpageheight}{11in} % DO NOT CHANGE THIS
%
% Keep the \pdfinfo as shown here. There's no need
% for you to add the /Title and /Author tags.
\pdfinfo{
/TemplateVersion (2026.1)
}
```

Preparing Your Paper

After the preamble above, you should prepare your paper as follows:

```
\begin{document}
\maketitle
\begin{abstract}
%...
\end{abstract}
```

If you want to add links to the paper's code, dataset(s), and extended version or similar this is the place to add them, within a *links* environment:

```
\begin{links}
  \link{Code}{https://aaai.org/example/guidelines}
  \link{Datasets}{https://aaai.org/example/datasets}
  \link{Extended version}{https://aaai.org/example}
```

```
\end{links}
```

You should then continue with the body of your paper. Your paper must conclude with the references, which should be inserted as follows:

```
% References and End of Paper
% These lines must be placed at the end of your paper
\bibliography{Bibliography-File}
\end{document}

\begin{document}\\
\maketitle\\
...\\
\bibliography{Bibliography-File}\\
\end{document}\\
```

Commands and Packages That May Not Be Used

There are a number of packages, commands, scripts, and macros that are incompatable with aaai2026.sty. The common ones are listed in tables 1 and 2. Generally, if a command, package, script, or macro alters floats, margins, fonts, sizing, linespacing, or the presentation of the references and citations, it is unacceptable. Note that negative vskip and vspace may not be used except in certain rare occurances, and may never be used around tables, figures, captions, sections, subsections, subsubsections, or references.

Page Breaks

For your final camera ready copy, you must not use any page break commands. References must flow directly after the text without breaks. Note that some conferences require references to be on a separate page during the review process. AAAI Press, however, does not require this condition for the final paper.

Paper Size, Margins, and Column Width

Papers must be formatted to print in two-column format on 8.5 x 11 inch US letter-sized paper. The margins must be exactly as follows:

- Top margin: .75 inches
- Left margin: .75 inches
- Right margin: .75 inches
- Bottom margin: 1.25 inches

The default paper size in most installations of LaTeX is A4. However, because we require that your electronic paper be formatted in US letter size, the preamble we have provided includes commands that alter the default to US letter size. Please note that using any other package to alter page size (such as, but not limited to the Geometry package) will result in your final paper being returned to you for correction.

Column Width and Margins. To ensure maximum readability, your paper must include two columns. Each column should be 3.3 inches wide (slightly more than 3.25 inches), with a .375 inch (.952 cm) gutter of white space between the two columns. The aaai2026.sty file will automatically create these columns for you.

Overlength Papers

If your paper is too long and you resort to formatting tricks to make it fit, it is quite likely that it will be returned to you. The best way to retain readability if the paper is overlength is to cut text, figures, or tables. There are a few acceptable ways to reduce paper size that don't affect readability. First, turn on \frenchspacing, which will reduce the space after periods. Next, move all your figures and tables to the top of the page. Consider removing less important portions of a figure. If you use \centering instead of \begin{center} in your figure environment, you can also buy some space. For mathematical environments, you may reduce fontsize **but not below 6.5 point**.

Commands that alter page layout are forbidden. These include \columnsep, \float, \topmargin, \topskip, \textheight, \textwidth, \oddsidemargin, and \evensizemargin (this list is not exhaustive). If you alter page layout, you will be required to pay the page fee. Other commands that are questionable and may cause your paper to be rejected include \parindent, and \parskip. Commands that alter the space between sections are forbidden. The title sec package is not allowed. Regardless of the above, if your paper is obviously "squeezed" it is not going to to be accepted. Options for reducing the length of a paper include reducing the size of your graphics, cutting text, or paying the extra page charge (if it is offered).

Type Font and Size

Your paper must be formatted in Times Roman or Nimbus. We will not accept papers formatted using Computer Modern or Palatino or some other font as the text or heading typeface. Sans serif, when used, should be Courier. Use Symbol or Lucida or Computer Modern for *mathematics only.*

Do not use type 3 fonts for any portion of your paper, including graphics. Type 3 bitmapped fonts are designed for fixed resolution printers. Most print at 300 dpi even if the printer resolution is 1200 dpi or higher. They also often cause high resolution imagesetter devices to crash. Consequently, AAAI will not accept electronic files containing obsolete type 3 fonts. Files containing those fonts (even in graphics) will be rejected. (Authors using blackboard symbols must avoid packages that use type 3 fonts.)

Fortunately, there are effective workarounds that will prevent your file from embedding type 3 bitmapped fonts. The easiest workaround is to use the required times, helvet, and courier packages with LaTeX2e. (Note that papers formatted in this way will still use Computer Modern for the mathematics. To make the math look good, you'll either have to use Symbol or Lucida, or you will need to install type 1 Computer Modern fonts — for more on these fonts, see the section "Obtaining Type 1 Computer Modern.")

If you are unsure if your paper contains type 3 fonts, view the PDF in Acrobat Reader. The Properties/Fonts window will display the font name, font type, and encoding properties of all the fonts in the document. If you are unsure if your graphics contain type 3 fonts (and they are PostScript or encapsulated PostScript documents), create PDF versions of them, and consult the properties window in Acrobat Reader.

\abovecaption	\abovedisplay	\addevensidemargin	\addsidemargin
\addtolength	\baselinestretch	\belowcaption	\belowdisplay
\break	\clearpage	\clip	\columnsep
\float	\input	\input	\linespread
\newpage	\pagebreak	\renewcommand	\setlength
\text height	\tiny	\top margin	\trim
\vskip{-	\vspace{-		

Table 1: Commands that must not be used

authblk	babel	cjk	dvips
epsf	epsfig	euler	float
fullpage	geometry	graphics	hyperref
layout	linespread	lmodern	maltepaper
navigator	pdfcomment	pgfplots	psfig
pstricks	t1enc	titlesec	tocbind
ulem			

Table 2: LaTeX style packages that must not be used.

The default size for your type must be ten-point with twelve-point leading (line spacing). Start all pages (except the first) directly under the top margin. (See the next section for instructions on formatting the title page.) Indent ten points when beginning a new paragraph, unless the paragraph begins directly below a heading or subheading.

Obtaining Type 1 Computer Modern for LaTeX. If you use Computer Modern for the mathematics in your paper (you cannot use it for the text) you may need to download type 1 Computer fonts. They are available without charge from the American Mathematical Society: http://www.ams.org/tex/type1-fonts.html.

Nonroman Fonts. If your paper includes symbols in other languages (such as, but not limited to, Arabic, Chinese, Hebrew, Japanese, Thai, Russian and other Cyrillic languages), you must restrict their use to bit-mapped figures.

Title and Authors

Your title must appear centered over both text columns in sixteen-point bold type (twenty-four point leading). The title must be written in Title Case according to the Chicago Manual of Style rules. The rules are a bit involved, but in general verbs (including short verbs like be, is, using, and go), nouns, adverbs, adjectives, and pronouns should be capitalized, (including both words in hyphenated terms), while articles, conjunctions, and prepositions are lower case unless they directly follow a colon or long dash. You can use the online tool https://titlecaseconverter.com/ to double-check the proper capitalization (select the "Chicago" style and mark the "Show explanations" checkbox).

Author's names should appear below the title of the paper, centered in twelve-point type (with fifteen point leading), along with affiliation(s) and complete address(es) (including electronic mail address if available) in nine-point roman type (the twelve point leading). You should begin the two-column format when you come to the abstract.

Formatting Author Information. Author information has to be set according to the following specification depending if you have one or more than one affiliation. You may not use a table nor may you employ the \authorblk.sty package. For one or several authors from the same institution, please separate them with commas and write all affiliation directly below (one affiliation per line) using the macros \author and \affiliations:

```
\author{
    Author 1, ..., Author n\\
}
\affiliations {
    Address line\\
    ... \\
    Address line\\
}
```

For authors from different institutions, use \textsuperscript {\rm x } to match authors and affiliations. Notice that there should not be any spaces between the author name and the superscript (and the comma should come after the superscripts).

```
\author{
    AuthorOne\equalcontrib\textsuperscript{\rm 1,\rm2},
    AuthorTwo\equalcontrib\textsuperscript{\rm 2},
    AuthorThree\textsuperscript{\rm 3},\\
    AuthorFour\textsuperscript{\rm 4},
    AuthorFive\textsuperscript{\rm 5}}
}
\affiliations {
    \textsuperscript{\rm 1}AffiliationOne,\\
    \textsuperscript{\rm 2}AffiliationTwo,\\
    \textsuperscript{\rm 3}AffiliationThree,\\
    \textsuperscript{\rm 4}AffiliationFour,\\
    \textsuperscript{\rm 5}AffiliationFive\\
    \{email, email\}@affiliation.com,
    email@affiliation.com,
    email@affiliation.com,
    email@affiliation.com
}
```

You can indicate that some authors contributed equally using the \equalcontrib command. This will add a marker after the author names and a footnote on the first page.

Note that you may want to break the author list for better visualization. You can achieve this using a simple line break (\\).

LaTeX Copyright Notice

The copyright notice automatically appears if you use aaai2026.sty. It has been hardcoded and may not be disabled.

Credits

Any credits to a sponsoring agency should appear in the acknowledgments section, unless the agency requires different placement. If it is necessary to include this information on the front page, use \thanks in either the \author or \title commands. For example:

> \title{Very Important Results in AI\thanks{This work is supported by everybody.}}

Multiple \thanks commands can be given. Each will result in a separate footnote indication in the author or title with the corresponding text at the botton of the first column of the document. Note that the \thanks command is fragile. You will need to use \protect.

Please do not include \pubnote commands in your document.

Abstract

Follow the example commands in this document for creation of your abstract. The command \begin{abstract} will automatically indent the text block. Please do not indent it further. Do not include references in your abstract!

Page Numbers

Do not print any page numbers on your paper. The use of \pagestyle is forbidden.

Text

The main body of the paper must be formatted in black, ten-point Times Roman with twelve-point leading (line spacing). You may not reduce font size or the linespacing. Commands that alter font size or line spacing (including, but not limited to baselinestretch, baselineshift, linespread, and others) are expressly forbidden. In addition, you may not use color in the text.

Citations

Citations within the text should include the author's last name and year, for example (Newell 1980). Append lower-case letters to the year in cases of ambiguity. Multiple authors should be treated as follows: (Feigenbaum and Engelmore 1988) or (Ford, Hayes, and Glymour 1992). In the case of four or more authors, list only the first author, followed by et al. (Ford et al. 1997).

Extracts

Long quotations and extracts should be indented ten points from the left and right margins.

> This is an example of an extract or quotation. Note the indent on both sides. Quotation marks are not necessary if you offset the text in a block like this, and properly identify and cite the quotation in the text.

Footnotes

Use footnotes judiciously, taking into account that they interrupt the reading of the text. When required, they should be consecutively numbered throughout with superscript Arabic numbers. Footnotes should appear at the bottom of the page, separated from the text by a blank line space and a thin, half-point rule.

Headings and Sections

When necessary, headings should be used to separate major sections of your paper. Remember, you are writing a short paper, not a lengthy book! An overabundance of headings will tend to make your paper look more like an outline than a paper. The aaai2026.sty package will create headings for you. Do not alter their size nor their spacing above or below.

Section Numbers. The use of section numbers in AAAI Press papers is optional. To use section numbers in LaTeX, uncomment the setcounter line in your document preamble and change the 0 to a 1. Section numbers should not be used in short poster papers and/or extended abstracts.

Section Headings. Sections should be arranged and headed as follows:

1. Main content sections
2. Appendices (optional)
3. Ethical Statement (optional, unnumbered)
4. Acknowledgements (optional, unnumbered)
5. References (unnumbered)

Appendices. Any appendices must appear after the main content. If your main sections are numbered, appendix sections must use letters instead of arabic numerals. In LaTeX you can use the `\appendix` command to achieve this effect and then use `\section{Heading}` normally for your appendix sections.

Ethical Statement. You can write a statement about the potential ethical impact of your work, including its broad societal implications, both positive and negative. If included, such statement must be written in an unnumbered section titled *Ethical Statement*.

Acknowledgments. The acknowledgments section, if included, appears right before the references and is headed "Acknowledgments". It must not be numbered even if other sections are (use `\section*{Acknowledgements}` in LaTeX). This section includes acknowledgments of help from associates and colleagues, credits to sponsoring agencies, financial support, and permission to publish. Please acknowledge other contributors, grant support, and so forth, in this section. Do not put acknowledgments in a footnote on the first page. If your grant agency requires acknowledgment of the grant on page 1, limit the footnote to the required statement, and put the remaining acknowledgments at the back. Please try to limit acknowledgments to no more than three sentences.

References. The references section should be labeled "References" and must appear at the very end of the paper (don't end the paper with references, and then put a figure by itself on the last page). A sample list of references is given later on in these instructions. Please use a consistent format for references. Poorly prepared or sloppy references reflect

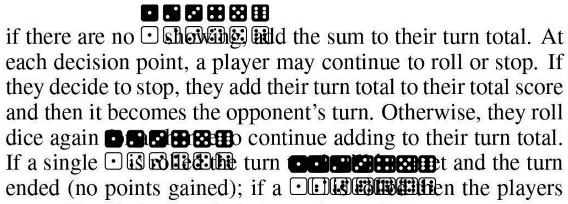

Figure 1: Using the trim and clip commands produces fragile layers that can result in disasters (like this one from an actual paper) when the color space is corrected or the PDF combined with others for the final proceedings. Crop your figures properly in a graphics program – not in LaTeX.

badly on the quality of your paper and your research. Please prepare complete and accurate citations.

Illustrations and Figures

Your paper must compile in PDFLATEX. Consequently, all your figures must be .jpg, .png, or .pdf. You may not use the .gif (the resolution is too low), .ps, or .eps file format for your figures.

Figures, drawings, tables, and photographs should be placed throughout the paper on the page (or the subsequent page) where they are first discussed. Do not group them together at the end of the paper. If placed at the top of the paper, illustrations may run across both columns. Figures must not invade the top, bottom, or side margin areas. Figures must be inserted using the \usepackage{graphicx}. Number figures sequentially, for example, figure 1, and so on. Do not use minipage to group figures.

If you normally create your figures using pgfplots, please create the figures first, and then import them as pdfs with proper bounding boxes, as the bounding and trim boxes created by pfgplots are fragile and not valid.

When you include your figures, you must crop them **outside** of LATEX. The command \includegraphics*[clip=true, viewport 0 0 10 10]... might result in a PDF that looks great, but the image is **not really cropped.** The full image can reappear (and obscure whatever it is overlapping) when page numbers are applied or color space is standardized. Figures 1, and 2 display some unwanted results that often occur.

If your paper includes illustrations that are not compatible with PDFTEX (such as .eps or .ps documents), you will need to convert them. The epstopdf package will usually work for eps files. You will need to convert your ps files to PDF in either case.

Figure Captions. The illustration number and caption must appear *under* the illustration. Labels and other text with the actual illustration must be at least nine-point type. However, the font and size of figure captions must be 10 point roman. Do not make them smaller, bold, or italic. (Individual words may be italicized if the context requires differentiation.)

Tables

Tables should be presented in 10 point roman type. If necessary, they may be altered to 9 point type. You must not use `\resizebox` or other commands that resize the entire table to make it smaller, because you can't control the final font size this way. If your table is too large you can use `\setlength{\tabcolsep}{1mm}` to compress the columns a bit or you can adapt the content (e.g.: reduce the decimal precision when presenting numbers, use shortened column titles, make some column duble-line to get it narrower).

Tables that do not fit in a single column must be placed across double columns. If your table won't fit within the margins even when spanning both columns and using the above techniques, you must split it in two separate tables.

Table Captions. The number and caption for your table must appear *under* (not above) the table. Additionally, the font and size of table captions must be 10 point roman and must be placed beneath the figure. Do not make them smaller, bold, or italic. (Individual words may be italicized if the context requires differentiation.)

Low-Resolution Bitmaps. You may not use low-resolution (such as 72 dpi) screen-dumps and GIF files—these files contain so few pixels that they are always blurry, and illegible when printed. If they are color, they will become an indecipherable mess when converted to black and white. This is always the case with gif files, which should never be used. The resolution of screen dumps can be increased by reducing the print size of the original file while retaining the same number of pixels. You can also enlarge files by manipulating them in software such as PhotoShop. Your figures should be 300 dpi when incorporated into your document.

LATEX Overflow. LATEX users please beware: LATEX will sometimes put portions of the figure or table or an equation in the margin. If this happens, you need to make the figure or table span both columns. If absolutely necessary, you may reduce the figure, or reformat the equation, or reconfigure the table. **Check your log file!** You must fix any overflow into the margin (that means no overfull boxes in LATEX). **Nothing is permitted to intrude into the margin or gutter.**

Using Color. Use of color is restricted to figures only. It must be WACG 2.0 compliant. (That is, the contrast ratio must be greater than 4.5:1 no matter the font size.) It must be CMYK, NOT RGB. It may never be used for any portion of the text of your paper. The archival version of your paper will be printed in black and white and grayscale. The web version must be readable by persons with disabilities. Consequently, because conversion to grayscale can cause undesirable effects (red changes to black, yellow can disappear, and so forth), we strongly suggest you avoid placing color figures in your document. If you do include color figures, you must (1) use the CMYK (not RGB) colorspace and (2) be mindful of readers who may happen to have trouble distinguishing colors. Your paper must be decipherable without using color for distinction.

Drawings. We suggest you use computer drawing software (such as Adobe Illustrator or, (if unavoidable), the

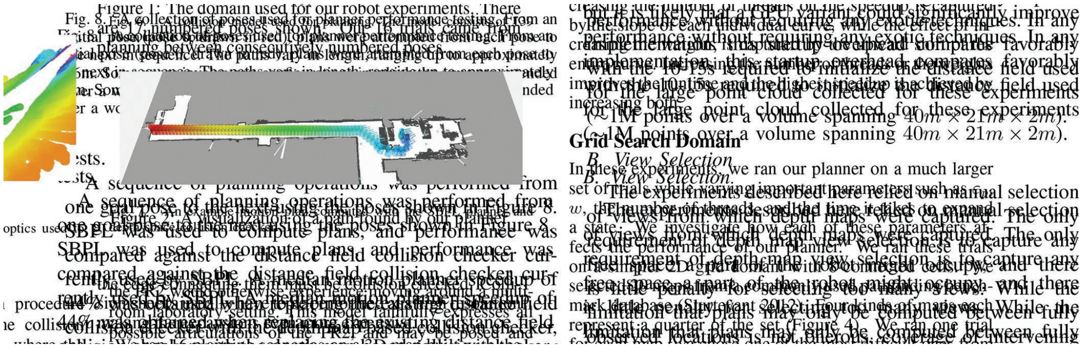

Figure 2: Adjusting the bounding box instead of actually removing the unwanted data resulted multiple layers in this paper. It also needlessly increased the PDF size. In this case, the size of the unwanted layer doubled the paper's size, and produced the following surprising results in final production. Crop your figures properly in a graphics program. Don't just alter the bounding box.

drawing tools in Microsoft Word) to create your illustrations. Do not use Microsoft Publisher. These illustrations will look best if all line widths are uniform (half- to two-point in size), and you do not create labels over shaded areas. Shading should be 133 lines per inch if possible. Use Times Roman or Helvetica for all figure call-outs. **Do not use hairline width lines** — be sure that the stroke width of all lines is at least .5 pt. Zero point lines will print on a laser printer, but will completely disappear on the high-resolution devices used by our printers.

Photographs and Images. Photographs and other images should be in grayscale (color photographs will not reproduce well; for example, red tones will reproduce as black, yellow may turn to white, and so forth) and set to a minimum of 300 dpi. Do not prescreen images.

Resizing Graphics. Resize your graphics **before** you include them with LaTeX. You may **not** use trim or clip options as part of your \includegraphics command. Resize the media box of your PDF using a graphics program instead.

Fonts in Your Illustrations. You must embed all fonts in your graphics before including them in your LaTeX document.

Algorithms. Algorithms and/or programs are a special kind of figures. Like all illustrations, they should appear floated to the top (preferably) or bottom of the page. However, their caption should appear in the header, left-justified and enclosed between horizontal lines, as shown in Algorithm 2. The algorithm body should be terminated with another horizontal line. It is up to the authors to decide whether to show line numbers or not, how to format comments, etc.

In LaTeX algorithms may be typeset using the `algorithm` and `algorithmic` packages, but you can also use one of the many other packages for the task.

Listings. Listings are much like algorithms and programs. They should also appear floated to the top (preferably) or bottom of the page. Listing captions should appear in the header, left-justified and enclosed between horizontal lines

Algorithm 2: Example algorithm

Input: Your algorithm's input
Parameter: Optional list of parameters
Output: Your algorithm's output

```
Let t = 0.
while condition do
    Do some action.
    if conditional then
        Perform task A.
    else
        Perform task B.
    end if
end while
return solution
```

Listing 1: Example listing `quicksort.hs`

```
quicksort :: Ord a => [a] -> [a]
quicksort []     = []
quicksort (p:xs) = (quicksort lesser) ++
      [p] ++ (quicksort greater)
  where
    lesser  = filter (< p) xs
    greater = filter (>= p) xs
```

as shown in Listing 1. Terminate the body with another horizontal line and avoid any background color. Line numbers, if included, must appear within the text column.

References

The AAAI style includes a set of definitions for use in formatting references with BibTeX. These definitions make the bibliography style fairly close to the ones specified in the Reference Examples appendix below. To use these definitions, you also need the BibTeX style file "aaai2026.bst," available in the AAAI Author Kit on the AAAI web site. Then, at the end of your paper but before \enddocument, you need to put the following lines:

\bibliography{bibfile1,bibfile2,...}

Please note that the aaai2026.sty class already sets the bibliographystyle for you, so you do not have to place any \bibliographystyle command in the document yourselves. The aaai2026.sty file is incompatible with the hyperref and navigator packages. If you use either, your references will be garbled and your paper will be returned to you.

References may be the same size as surrounding text. However, in this section (only), you may reduce the size to *small* (9pt) if your paper exceeds the allowable number of pages. Making it any smaller than 9 point with 10 point linespacing, however, is not allowed.

The list of files in the \bibliography command should be the names of your BibTeX source files (that is, the .bib files referenced in your paper).

The following commands are available for your use in citing references:

cite: Cites the given reference(s) with a full citation. This appears as "(Author Year)" for one reference, or "(Author Year; Author Year)" for multiple references.

shortcite: Cites the given reference(s) with just the year. This appears as "(Year)" for one reference, or "(Year; Year)" for multiple references.

citeauthor: Cites the given reference(s) with just the author name(s) and no parentheses.

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You may also use any of the *natbib* citation commands.

Proofreading Your PDF

Please check all the pages of your PDF file. The most commonly forgotten element is the acknowledgements — especially the correct grant number. Authors also commonly forget to add the metadata to the source, use the wrong reference style file, or don't follow the capitalization rules or comma placement for their author-title information properly. A final common problem is text (expecially equations) that runs into the margin. You will need to fix these common errors before submitting your file.

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Additional Resources

LaTeX is a difficult program to master. If you've used that software, and this document didn't help or some items were not explained clearly, we recommend you read Michael Shell's excellent document (testflow doc.txt V1.0a 2002/08/13) about obtaining correct PS/PDF output on LaTeX systems. (It was written for another purpose, but it has general application as well). It is available at www.ctan.org in the tex-archive.

Reference Examples

Formatted bibliographies should look like the following examples. You should use BibTeX to generate the references. Missing fields are unacceptable when compiling references, and usually indicate that you are using the wrong type of entry (BibTeX class).

Book with multiple authors Use the `@book` class.

Engelmore, R.; and Morgan, A., eds. 1986. *Blackboard Systems*. Reading, Mass.: Addison-Wesley.

Journal and magazine articles Use the `@article` class.

Robinson, A. L. 1980. New Ways to Make Microcircuits Smaller. *Science*, 208(4447): 1019–1022.

Hasling, D. W.; Clancey, W. J.; and Rennels, G. 1984. Strategic explanations for a diagnostic consultation system. *International Journal of Man-Machine Studies*, 20(1): 3–19.

Proceedings paper published by a society, press or publisher Use the `@inproceedings` class. You may abbreviate the *booktitle* field, but make sure that the conference edition is clear.

Clancey, W. J. 1984. Classification Problem Solving. In *Proceedings of the Fourth National Conference on Artificial Intelligence*, 45–54. Menlo Park, Calif.: AAAI Press.

Clancey, W. J. 1983. Communication, Simulation, and Intelligent Agents: Implications of Personal Intelligent Machines for Medical Education. In *Proceedings of the Eighth International Joint Conference on Artificial Intelligence (IJCAI-83)*, 556–560. Menlo Park, Calif: IJCAI Organization.

University technical report Use the `@techreport` class.

Rice, J. 1986. Poligon: A System for Parallel Problem Solving. Technical Report KSL-86-19, Dept. of Computer Science, Stanford Univ.

Dissertation or thesis Use the `@phdthesis` class.

Clancey, W. J. 1979. *Transfer of Rule-Based Expertise through a Tutorial Dialogue*. Ph.D. diss., Dept. of Computer Science, Stanford Univ., Stanford, Calif.

Forthcoming publication Use the `@misc` class with a `note="Forthcoming"` annotation.

```
@misc(key,
  [...]
  note="Forthcoming",
)
```

Clancey, W. J. 2021. The Engineering of Qualitative Models. Forthcoming.

ArXiv paper Fetch the BibTeX entry from the "Export Bibtex Citation" link in the arXiv website. Notice it uses the `@misc` class instead of the `@article` one, and that it includes the `eprint` and `archivePrefix` keys.

```
@misc(key,
  [...]
  eprint="xxxx.yyyy",
  archivePrefix="arXiv",
)
```

Vaswani, A.; Shazeer, N.; Parmar, N.; Uszkoreit, J.; Jones, L.; Gomez, A. N.; Kaiser, L.; and Polosukhin, I. 2017. Attention Is All You Need. arXiv:1706.03762.

Website or online resource Use the `@misc` class. Add the url in the `howpublished` field and the date of access in the `note` field:

```
@misc(key,
  [...]
  howpublished="\url{http://...}",
  note="Accessed: YYYY-mm-dd",
)
```

NASA. 2015. Pluto: The 'Other' Red Planet. https://www.nasa.gov/nh/pluto-the-other-red-planet. Accessed: 2018-12-06.

For the most up to date version of the AAAI reference style, please consult the *AI Magazine* Author Guidelines at https://aaai.org/ojs/index.php/aimagazine/about/submissions#authorGuidelines

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Thank you for reading these instructions carefully. We look forward to receiving your electronic files!

References

Clancey, W. J. 1979. *Transfer of Rule-Based Expertise through a Tutorial Dialogue*. Ph.D. diss., Dept. of Computer Science, Stanford Univ., Stanford, Calif.

Clancey, W. J. 1983. Communication, Simulation, and Intelligent Agents: Implications of Personal Intelligent Machines for Medical Education. In *Proceedings of the Eighth International Joint Conference on Artificial Intelligence (IJCAI-83)*, 556–560. Menlo Park, Calif: IJCAI Organization.

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NASA. 2015. Pluto: The 'Other' Red Planet. https://www.nasa.gov/nh/pluto-the-other-red-planet. Accessed: 2018-12-06.

Rice, J. 1986. Poligon: A System for Parallel Problem Solving. Technical Report KSL-86-19, Dept. of Computer Science, Stanford Univ.

Robinson, A. L. 1980. New Ways to Make Microcircuits Smaller. *Science*, 208(4447): 1019–1022.

Vaswani, A.; Shazeer, N.; Parmar, N.; Uszkoreit, J.; Jones, L.; Gomez, A. N.; Kaiser, L.; and Polosukhin, I. 2017. Attention Is All You Need. arXiv:1706.03762.